VISUAL ANALYTICS:
EXPLORATORY DATA ANALYSIS (EDA)

AB MOSCA (THEY/THEM)

TODAY'S PLAN

- HW notes
- Motivation
- Define EDA
- Techniques
- Practice

HW NOTES

- Starter code for coding assignments will be on Github
  - Ex. https://github.com/amosca01/SDS-CSC-235-hw02

- **Fork the starter code repo** and then work out of your fork

- You can submit on Gradescope by linking your repository once you've pushed your final submission

MOTIVATION

- TMDB 5000 Movie Dataset: https://www.kaggle.com/datasets/tmdb/tmdb-movie-metadata?resource=download&select=tmdb_5000_movies.csv

- What would you do if someone handed you this dataset and asked you to analyze it?

DEFINITION

- Exploratory Data Analysis (EDA) is an approach of analyzing datasets that focuses on summarizing their main characteristics.
- Main objectives:
  - Enable unexpected discoveries in the data
  - Suggest hypotheses
  - Assess modeling assumptions
  - Support selection of appropriate modeling tools
  - Provide a basis for further exploration

John Tukey, Exploratory Data Alaysis, 1977. https://en.wikipedia.org/wiki/Exploratory_data_analysis#cite_note-Tukey1977-6

EDA TECHNIQUES

- Five number summary (for quantitative variables)

```
    revenue                runtime
 Min.   :0.000e+00   Min.   :  0.0
 1st Qu.:0.000e+00   1st Qu.: 94.0
 Median :1.917e+07   Median :103.0
 Mean   :8.226e+07   Mean   :106.9
 3rd Qu.:9.292e+07   3rd Qu.:118.0
 Max.   :2.788e+09   Max.   :338.0
                     NA's   :2
```

```
  vote_average     vote_count
 Min.   : 0.000   Min.   :    0.0
 1st Qu.: 5.600   1st Qu.:   54.0
 Median : 6.200   Median :  235.0
 Mean   : 6.092   Mean   :  690.2
 3rd Qu.: 6.800   3rd Qu.:  737.0
 Max.   :10.000   Max.   :13752.0
```

EDA TECHNIQUES

- Frequency Tables (categorical / ordinal variables)

```
> table(ds$status)

Post Production        Released        Rumored 
              3            4795              5 
```

```
> table(ds$original_language)

  af   ar   cn   cs   da   de   el   en   es   fa   fr   he   hi   hu   id   is   it   ja   ko   ky   nb   nl   no   pl 
   1    2   12    2    7   27    1 4505   32    4   70    3   19    1    2    1   14   16   11    1    1    4    1    1 
  ps   pt   ro   ru   sl   sv   ta   te   th   tr   vi   xx   zh 
   1    9    2   11    1    5    2    1    3    1    1    1   27 
```

EDA TECHNIQUES

- Frequency Tables (categorical / ordinal variables)

```
> table(ds$genres)
```

```
> table(ds$Genre1)

         "Action"       "Adventure"       "Animation"          "Comedy"           "Crime"     "Documentary"
              754               339               123              1042               195                89
          "Drama"          "Family"         "Fantasy"         "Foreign"         "History"          "Horror"
             1207                56               117                 2                25               300
          "Music"         "Mystery"         "Romance" "Science Fiction"        "Thriller"        "TV Movie"
               34                41               106                96               194                 4
            "War"         "Western"              #N/A
               24                27                28
```

[{"id": 10402, "name": "Music"}, {"id": 18, "name": "Drama"}, {"id": 10749, "name": "Romance"}]

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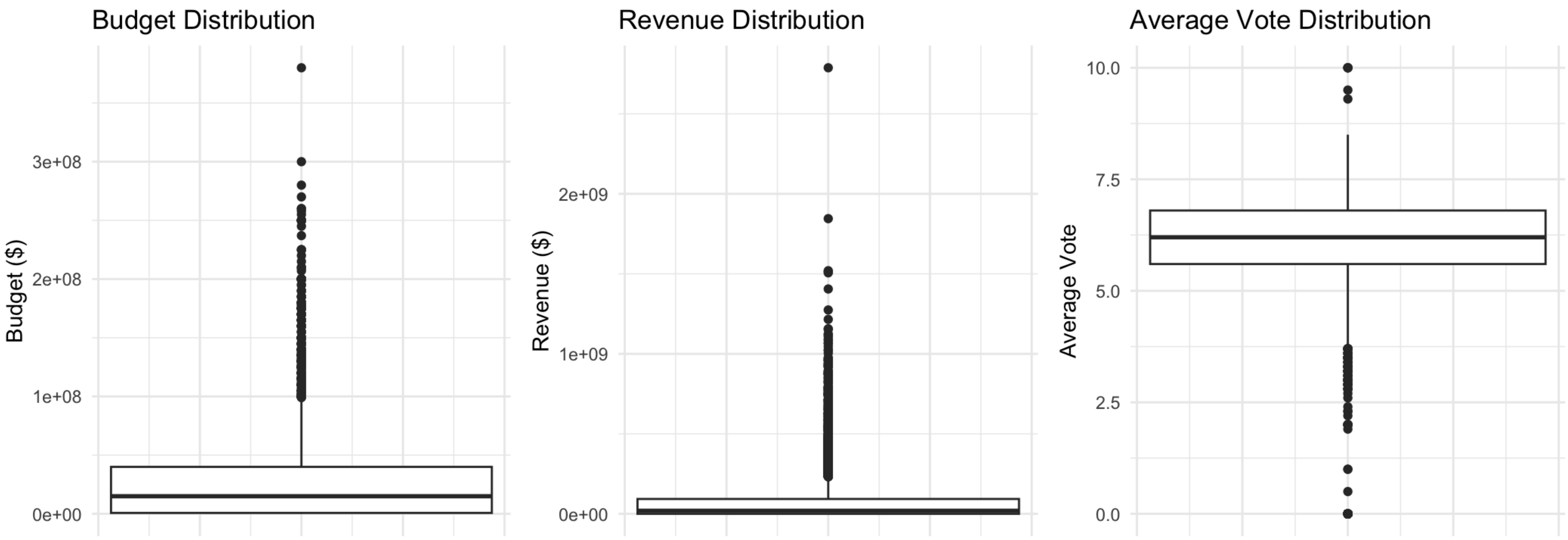
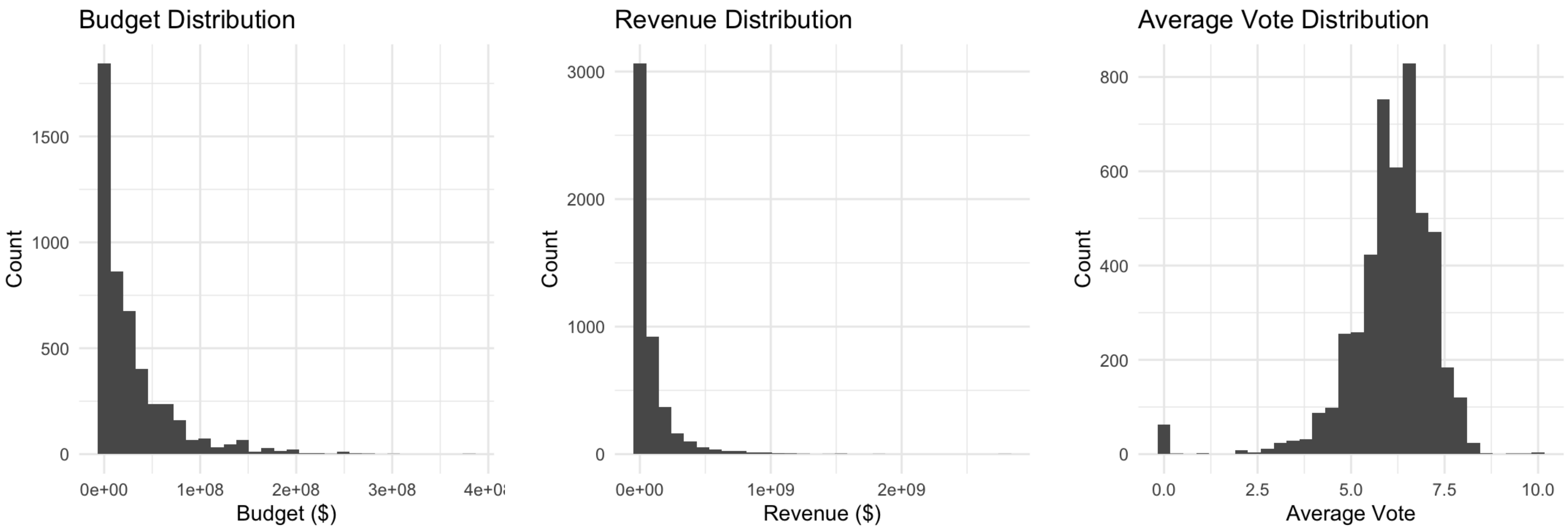
EDA TECHNIQUES

- Distributions (quantitative variables)



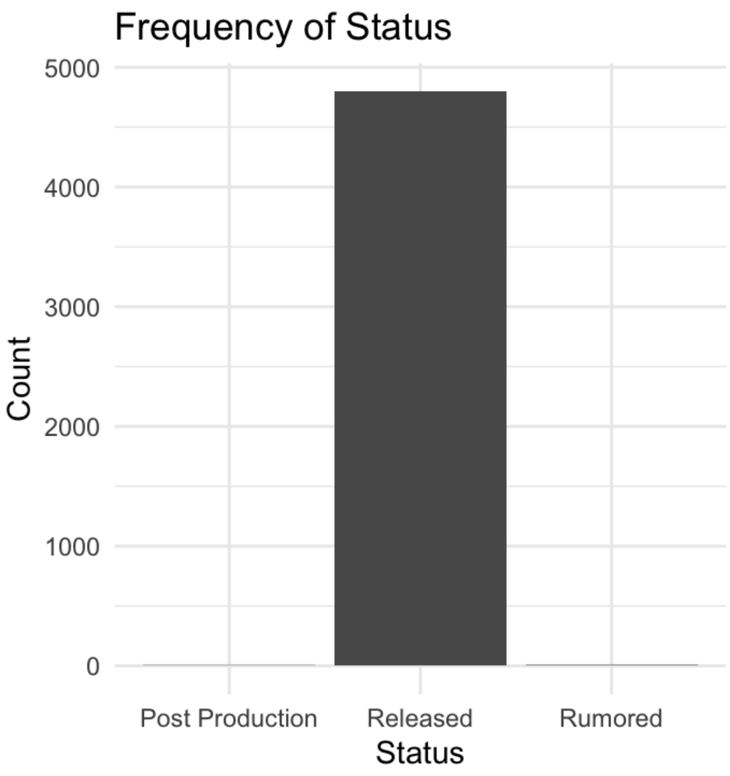
EDA TECHNIQUES

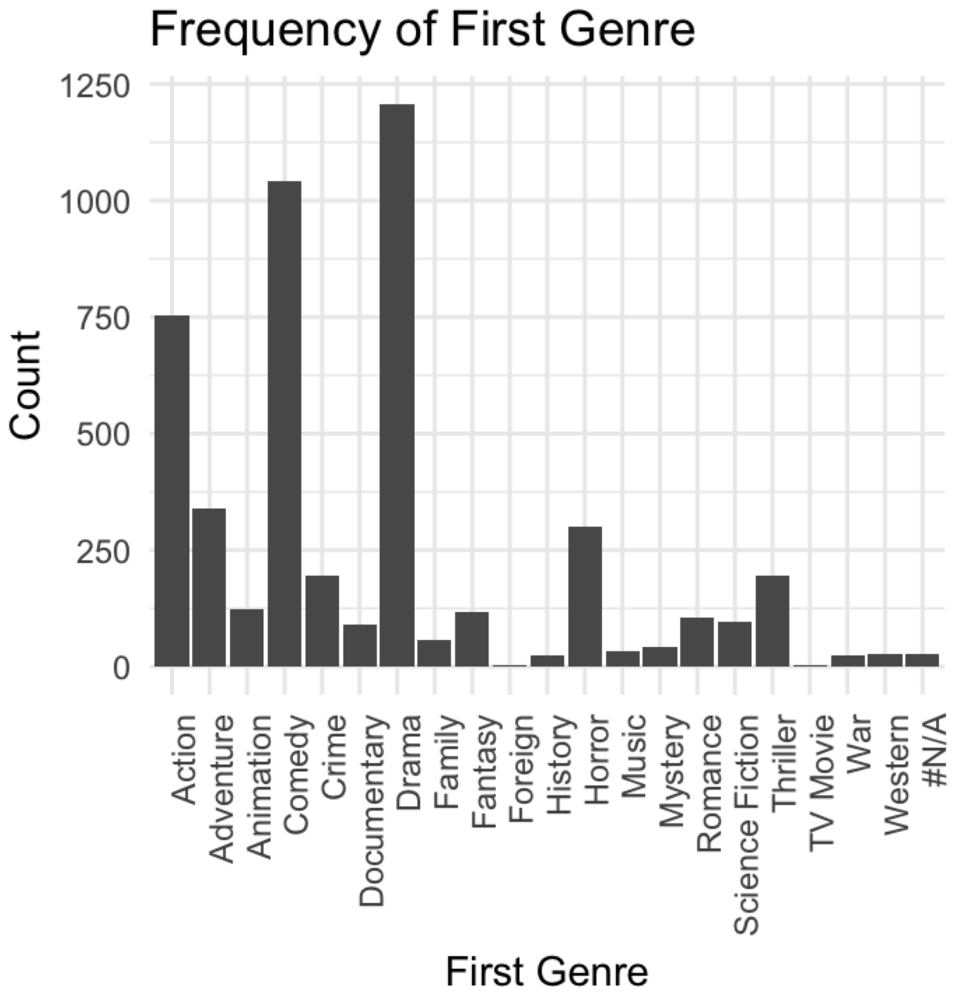
- Distributions (quantitative variables)



EDA TECHNIQUES

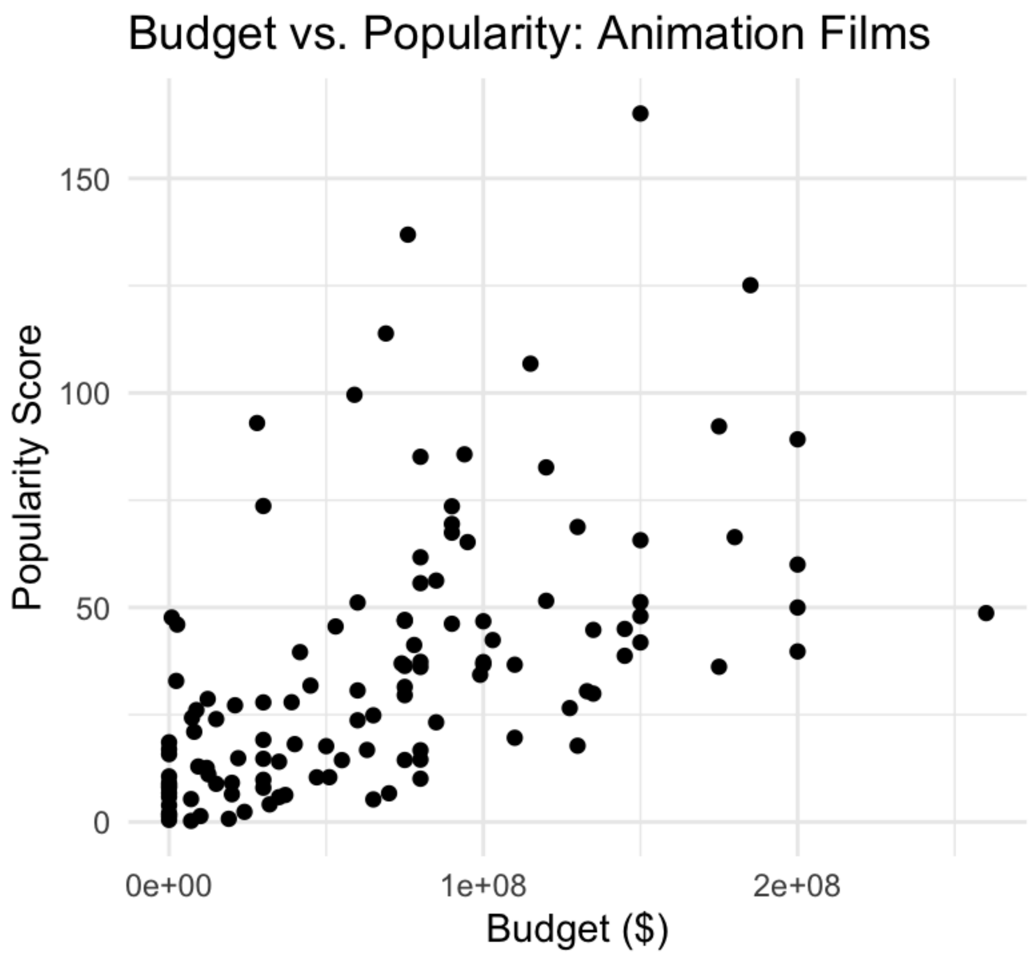
- Frequency (categorical / ordinal variables)





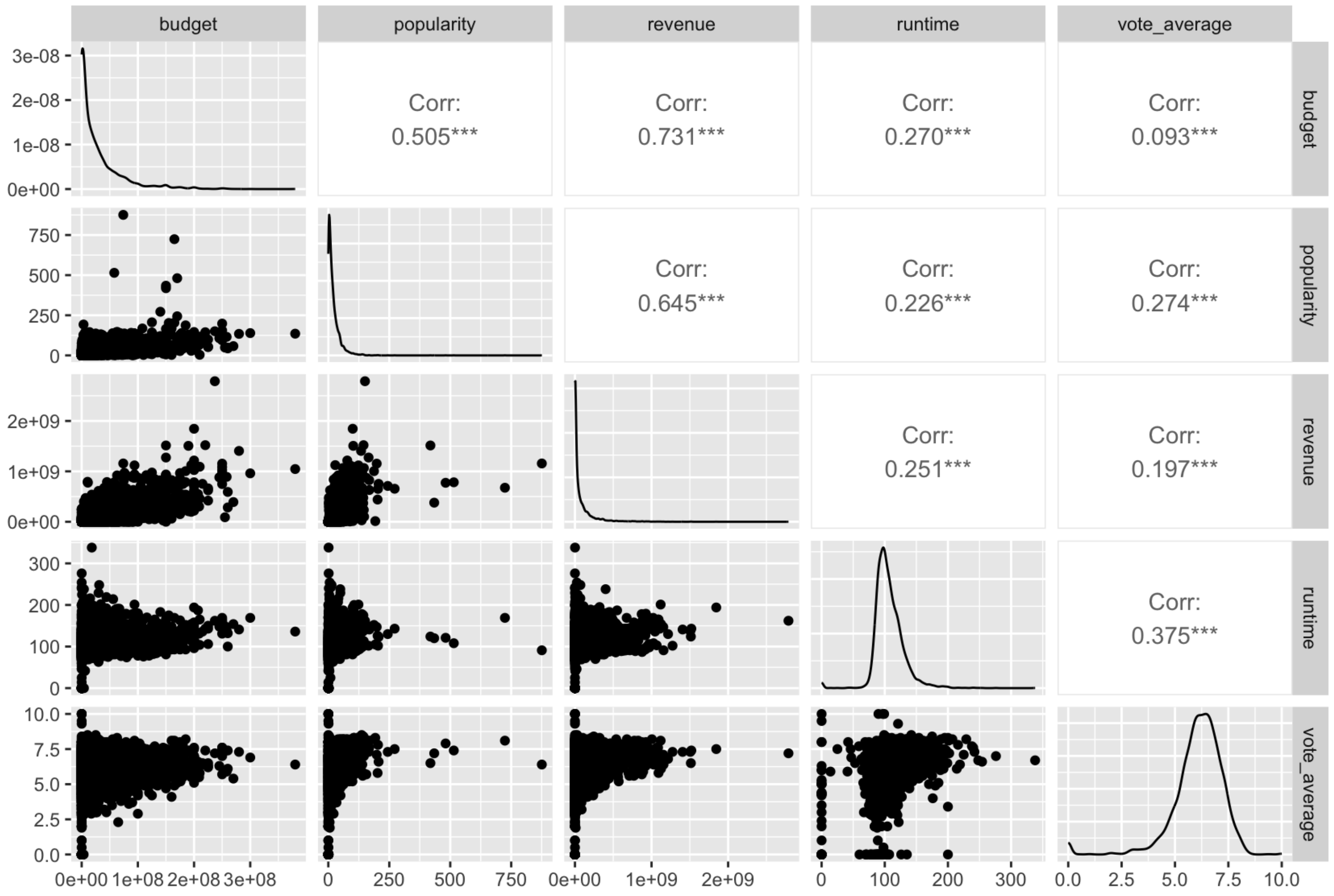
EDA TECHNIQUES

- Correlations (between quantitative variables)



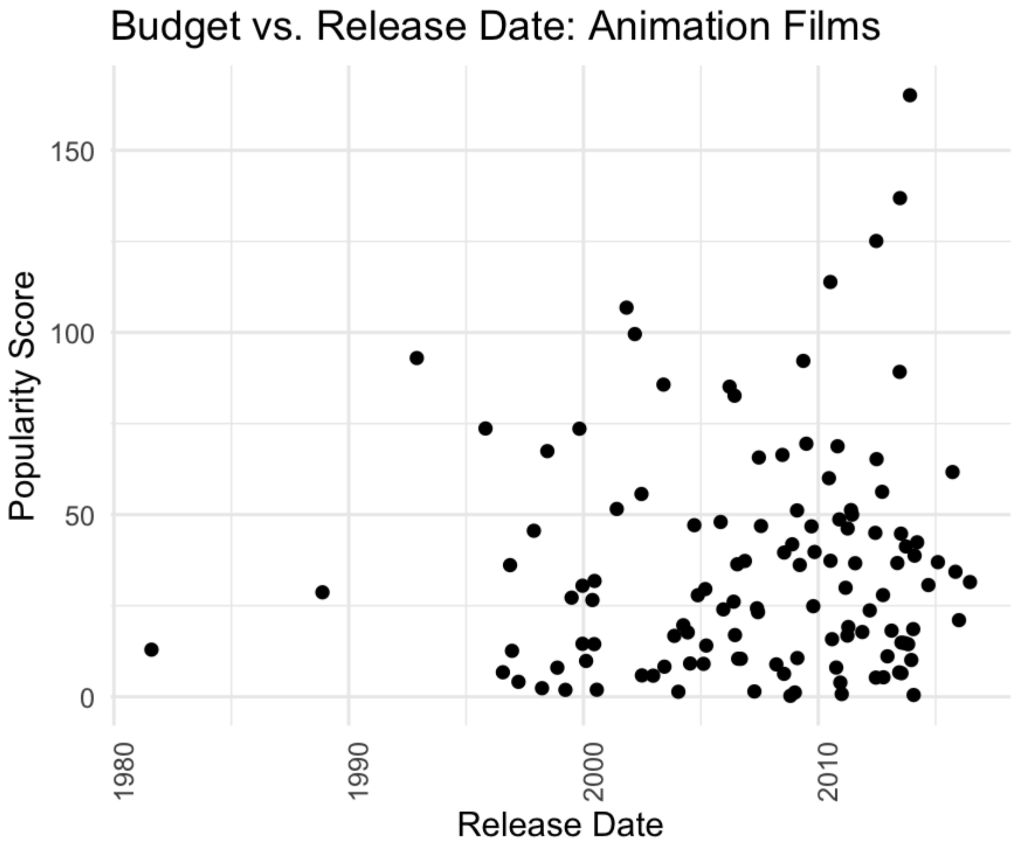
EDA TECHNIQUES

- Correlations (between many quantitative variables)



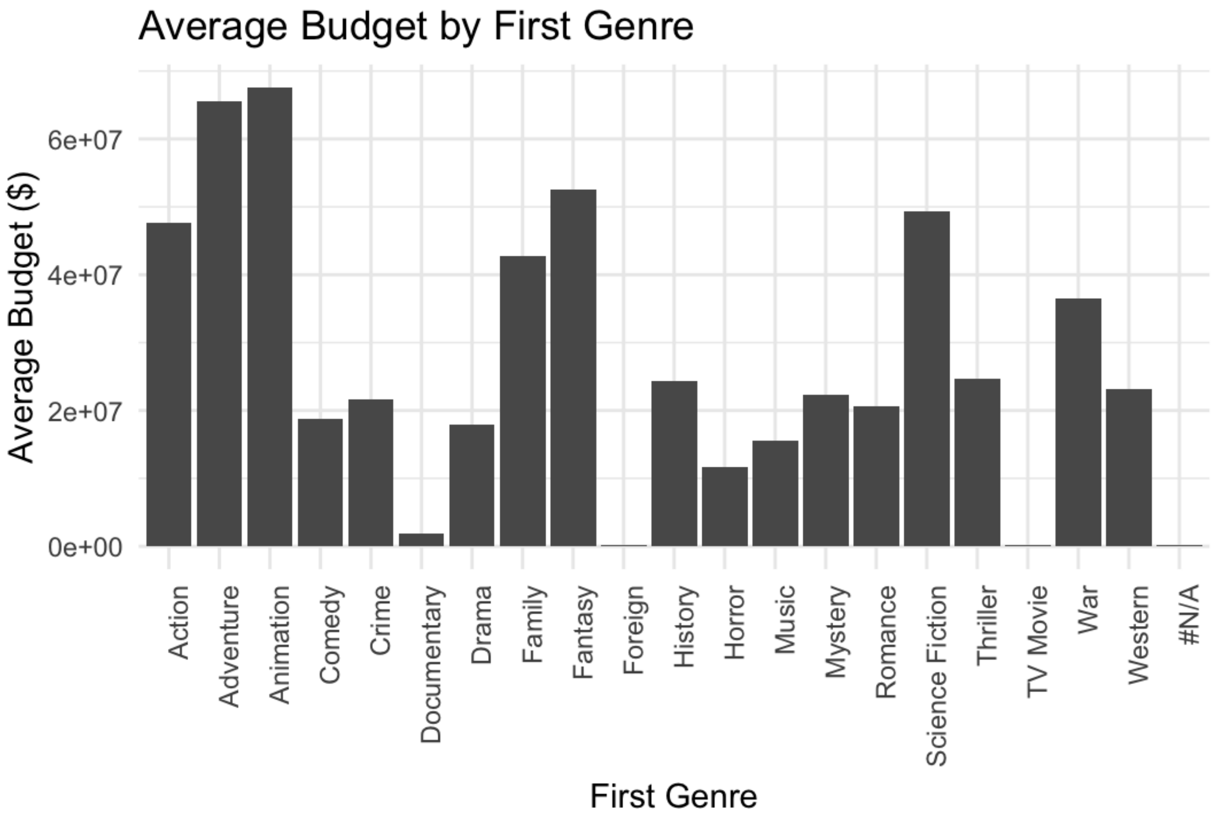
EDA TECHNIQUES

- Correlations (between quantitative / ordinal and quantitative variables)



EDA TECHNIQUES

- Correlations (between quantitative / categorical variables)



PRACTICE

- Data: Real Estate Sales 2001-2023 GL
  - https://catalog.data.gov/dataset/real-estate-sales-2001-2018
- Use whatever tool you like (ex. R, python) to perform EDA on the dataset
- As you perform EDA, do any necessary data cleaning / wrangling
- Identify:
  - Distributions of variables
  - Outliers
  - Correlations
- Brainstorm:
  - Three specific analysis questions you could answer with this dataset